

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I SHAKTHI /192312538 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Shokthi



1. Question 24 - CSP Exam Timetable

Schedule AI, ML, DBMS and OS into Monday 10 AM, Monday 2 PM and Tuesday 10 AM
Constraints AI and ML cannot share a slot, DBMS after AI, OS not Monday 2 PM, ML before OS

Use backtracking and show every assignment

Subject	Domain	Assigned Slot
AI	Mon10/Mon2/Tue10	
ML	Mon10/Mon2/Tue10	
DBMS	Mon10/Mon2/Tue10	
OS	Mon10/Mon2/Tue10	

2. Question 25 - Search Strategy Comparison

Consider a graph with S->A,B, A->C,D, B->E, C->G, D->G, E->G.

Perform BFS, DFS, UCS, Greedy Best-First Search and A* separately. Complete the comparison table

Algorithm	Expansion Order	Path	Cost	Complete?	Optimal?
1					
2					
3					
4					

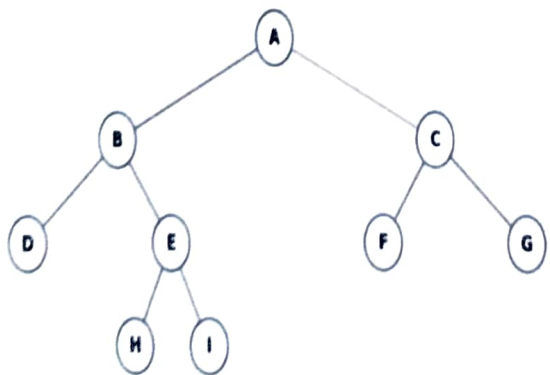
Algorithm	Expansion Order	Path	Cost	Complete?	Optimal?
5					

3. Question 1 - Breadth-First Search (BFS)

Consider the following graph.

Task: Starting from node A, perform Breadth-First Search (BFS).

Complete the table below.



Iteration	Queue Before	Expanded	Queue After
1			
2			
3			
4			
5			

A) Subjects :-

AI, ML, DBMS, OS

Slots :-

Monday 10AM, Monday 2PM, Tuesday 10AM

Constraints :-

⇒ AI and ML Cannot share a slot

⇒ DBMS must be after AI.

⇒ OS Cannot be Monday 2PM.

⇒ ML must be before OS.

Back tracking - Every Valid Assignment

Assignment	AI	ML	DBMS	OS
1.	Mon 10AM	Mon 2PM	Mon 2PM	Tue 10AM
2.	Mon 10AM	Mon 2PM	Tue 10AM	Tue 10AM
3.	Mon 2PM	Mon 10AM	Tue 10AM	Tue 10AM

Total Valid assignments = 3

2. A) Given graph:

$\Rightarrow S \rightarrow A, B$

$\Rightarrow A \rightarrow C, D$

$\Rightarrow B \rightarrow E$

$\Rightarrow C \rightarrow G$

$\Rightarrow D \rightarrow G$

$\Rightarrow E \rightarrow G$

Assume unit edge cost because no edge costs are given. Also, since no heuristic values are provided, for A^* we assume $h(n) = 0$.

Algorithm	Expansion order	Path to G	Cost	Complete	Optimal
BFS	S, A, B, C, D, E, G	S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G	3	Yes	Yes
DFS	S, A, C, G	S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G	3	Yes	No*
UCS	S, A, B, C, D, E, G	S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G	3	Yes	Yes
Greedy Best-first	S, A, C, G**	S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G	3	Yes	No*
A*	S, A, B, C, D, E, G**	S $\rightarrow$ A $\rightarrow$ C $\rightarrow$ G	3	Yes	Yes

$\Rightarrow$ DFS and Greedy are not guaranteed to find an optimal path in general, even though they give the optimal path in this particular graph

a) Graph:



Starting node = A

BFS Explanation order

A → B → C → D → E → F → G → H → I

Queue Table

Iteration	Queue Before	Expanded	Queue After
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G, H, I
6	F, G, H, I	F	G, H, I
7	G, H, I	G	H, I
8	H, I	H	I
9	I	I	Empty

BFS transversal = A, B, C, D, E, F, G, H, I

⇒ These orders requires a heuristic / tie-breaking assumption. No heuristic values are shown in the question.

Simple Explanation:-

⇒ BFS:-

$S \rightarrow A, B \rightarrow C, D, E \rightarrow G$

Path = $S-A-C-G$, Cost = 3

⇒ DFS:-

$S \rightarrow A \rightarrow C \rightarrow G$

Path = $S-A-C-G$, Cost = 3

UCS:-

All edges have Cost 1, So UCS behaves like BFS

Path = $S-A-C-G$, Cost = 3

Greedy:-

chooses the node that appears closest to G according to the heuristic. Since no heuristic is given, exact expansion order cannot be uniquely determined

A*:

uses:

$$f(n) = g(n) + h(n)$$

⇒ Since no heuristic is provided, assuming $h(n) = 0$, A* becomes equivalent to UCS.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

20 + 28 + 18 + 18 + 9 = 93